

High Performance Inkjet-Printed Quantum-Dot Light-Emitting Diodes with High Operational Stability

Siqi Jia, Haodong Tang, Jingrui Ma, Shihao Ding, Xiangwei Qu, Bing Xu, Zhenghui Wu, Guangyu Li, Pai Liu,* Kai Wang,* and Xiao Wei Sun*

Improving the stability of inkjet-printed quantum dot light emitting diodes (QLEDs) is critical for the technology to become commercially viable. The major obstacle is the compromise between the printability of the ink system and the functionality of the carrier transport layers. Here, a ternary ink system consisting of octane, 1-cyclohexyl-ethanol, and *n*-butyl acetate is reported, which solves the erosion between the printed quantum dot ink and the underneath hole transport layer. A gradient vacuum post-treatment is developed to accompany the ternary ink system with gradient vacuum pressures, which is helpful in forming a uniform printing layer. Based on both technologies, the inkjet-printed R/G/B QLEDs are fabricated with high resolution patterns, showing high efficiencies and stabilities. The external quantum efficiency of R/G/B devices is 19.3%, 18.0%, and 4.4%, respectively. Correspondingly, the half operating lifetime is up to 25 178 h @ 1000 cd m⁻², 20 655 h @ 1000 cd m⁻², and 46 h @ 100 cd m⁻², respectively. The improvements in the ink engineering and post-treatment in this study have taken the efficiency and stability of the devices to a higher level and confirm the application prospects of printed QLEDs in the display industry.

1. Introduction

Quantum dot light-emitting diode (QLED) is regarded as a mainstream candidate for next generation display applications, owing to their outstanding color purity, wide color gamut, and cost-effective fabrication.^[1–5] During the last two decades, the performance of QLED has made great progress following the continuous refinement of quantum dot (QD) synthesis and device development.^[6–9] However, most of the high performance QLEDs were fabricated by the spin coating (SC) technology, which is only available for small area devices in the laboratory.^[10–12] Since inkjet printing (IJP) is introduced into optoelectronics fabrication, there are many reports on the printed light emitting diodes for flexible and transparent display, which received great attention from both the academy and industry.^[13–15]

However, the poor efficiency and stability of IJP QLED compared with SC QLED, impede its further development into practical display panel processing.^[14,16–18] Therefore, many attempts have been made in promoting the efficiency of printed devices, such as by altering ligands of QDs to improve their performance,^[8,14,19] developing the suitable solvent system,^[16,20] and novel post-treatments.^[21] Recently, the efficiency and stability of red emitting IJP QLED were improved with an external quantum efficiency (EQE) over 16%, and half brightness @ 100 cd m⁻² lifetime of more than one million hours by exchanging the ligands from OA to Zn(OA)₂, to promote the charge injection.^[14] Different from that the QD solution spread out under the centrifugal force in SC process, the ink droplet in IJP process will drop on the substrate and then concentrated at the edge of the droplet as it dries up, which leads to the coffee ring effect.^[10,22,23] The coffee ring serves as a pattern left by a puddle of particle-laden liquid, which can lead to a serious rough surface.^[24] Meanwhile, a smooth, compact, and pinhole-free emitting QD layer is crucial for high efficiency and long lifetime QLED.^[5,20,25] Therefore, there is still an efficiency gap between the IJP and SC devices, which mainly results from the unideal surface of the printed layer.^[23,25,26] For this issue, it has been reported that mixtures of low vapor pressure and high vapor pressure solvents like alkane solvent and phenyl-cyclohexane (CHB) can suppress the coffee ring effect, which has been widely used in IJP -QLED.^[14,17,26] A greater solvent

S. Jia, H. Tang, J. Ma, S. Ding, X. Qu, B. Xu, Z. Wu, P. Liu, K. Wang, X. W. Sun
Key Laboratory of Energy Conversion and Storage Technologies
(Southern University of Science and Technology)
Ministry of Education
and Department of Electrical and Electronic Engineering
Southern University of Science and Technology
Shenzhen 518055, China
E-mail: liup7@sustech.edu.cn; wangk@sustech.edu.cn;
sunxw@sustech.edu.cn

S. Jia, H. Tang, J. Ma, X. Qu, Z. Wu, P. Liu, K. Wang, X. W. Sun
Guangdong University Key Laboratory for Advanced Quantum Dot
Displays and Lighting
Guangdong-Hong Kong-Macao Joint Laboratory for
Photonic-Thermal-Electrical Energy Materials and Devices
Shenzhen Key Laboratory for Advanced Quantum Dot Displays and Lighting
and Department of Electrical and Electronic Engineering
Southern University of Science and Technology
Shenzhen 518055, China

S. Jia, G. Li
Department of Materials Science and Engineering
Jilin University
Changchun 130025, China

B. Xu, X. W. Sun
Shenzhen Planck Innovation Technologies Co. Ltd.
Shenzhen 518055, China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adom.202101069>.

DOI: 10.1002/adom.202101069

evaporation flux can weaken the coffee ring effect to form a more uniform layer.^[21,28] Therefore, the post-treatment after IJP also plays a pivotal role in forming a uniform QD layer.

Besides, high vapor pressure solvent such as hexane evaporates quickly even before leaving the nozzle, which may lead to nozzle clogging. Moreover, traditional solvent can erode the underneath layer (generally the hole transport layer (HTL)).^[16,17,27,29] The limited choice of solvent is one of the main obstacles that prevent the wide adoption of the IJP OLED/QLED.^[15,26,30] For example, as a common HTL layer, the poly[(9,9-dioctylfluorenyl-2,7-diyl)-alt-(4,4'-(N-(4-butylphenyl) TFB) layer can be seriously damaged by the reported inks, because the main solvent CHB with the benzene group can dissolve TFB easily.^[26,30,31] Recent studies propose some strategies to overcome this, which include designing the crosslinkable HTL materials,^[30–32] or using anticorrosive HTL material like poly (N-vinyl carbazole) (PVK).^[26] However, compared to these HTL materials, TFB is more suitable for HTL layer in QLEDs with long operational lifetime due to its inherent electrostability.^[8,19,33] Thus, while the efficiency of IJP QLED was improved by these strategies, the operational lifetime of devices is compromised, so that they cannot meet the requirement of the industry. Therefore, designing a novel ink system compatible with a wide range of HTL materials is crucial to the highly efficient and stable IJP-QLED.

In summary, a suitable QD ink should fulfill the following: 1) the good solubility of QD; 2) good printability and film formation; 3) no-erosion to the TFB layer. Based on the above three requirements, we designed a novel ternary QD ink (TQ-ink) system which not only suppresses the coffee ring effect but also do not damage the TFB layer. Correspondingly, a three-stage gradient vacuum post-treatment (GVP) method is adopted to remove the ternary ink subsequently, which can further improve the quality of the printed layer. Based on the TQ-ink system and GVP technology, we printed a uniform layer of QD on TFB with a roughness of less than 1 nm. The ink system is orthogonal to the TFB layer; there is no damage to the TFB. Hence, the leakage current is significantly reduced and charge injection greatly enhanced, resulting in the excellent performance of the printed devices. The red IJP-QLED exhibited a maximum luminance of 104 679 cd m⁻², a maximum EQE of 19.3%, and a half brightness lifetime (T₅₀) up to 25, 178 h @ 1000 cd m⁻². Meanwhile, the green and blue IJP-QLEDs using the same TQ-ink system and GVP method also show outstanding EQE (18.0% and 4.4%) and operational stability (lifetime of 20, 655 h @ 1000 cd m⁻² and 46 h @ 100 cd m⁻²). The lifetimes and efficiencies of RGB IJP-QLED are the highest in printed devices reported so far.

2. Result and Discussion

The TQ-ink formula composed of the main solvent of QD which normally possesses a compound of high vapor pressure and low viscosity (<1 cP), such as octane (viscosity of 0.51 cP), and a portion of thickening agent with low vapor pressure to adjust the viscosity of the ink, 1-cyclohexyl-ethanol (viscosity of 2.5 cP) in this case. However, these two solvents show a large gap in their vapor pressure (octane: 14 mmHg; 1-cyclohexyl-ethanol:

0.15 mmHg at 20 °C), which will cause a completely discrete evaporation process. Though both solvents start to evaporate simultaneously, the evaporation of octane will be first completed, after which it becomes a single solvent (1-cyclohexyl-ethanol) evaporation process, which will prone to the coffee ring generation.^[24] Therefore, another solvent (*n*-butyl acetate) with intermediate vapor pressure (11.5 mmHg) was added to reconcile the two components. A control ink was also made by choosing from decane and CHB which are frequently chosen as QD ink components in literature.^[14,17] Tables S1 and S2 (Supporting Information) summarized the physical properties of the TQ-ink system and the control ink, such as viscosity and vapor pressure at room temperature.

A serial of inks (TQ inks and control ink) was prepared with a QD concentration of 15 mg mL⁻¹, and viscosity and surface tension as a function of solvent composition are tabulated in Table S2 (Supporting Information). Besides, as Figure S1 (Supporting Information) shows, the both ink systems have no effect on the PL decay and PLQY, indicating that the QD dispersed well in these ink systems. To evaluate a printable fluid and the interface rupture during droplet formation, the inverse of Ohnesorge number ($Z = 1/Oh$) is given by: $Z = \sqrt{\rho a \gamma} / \mu$, where ρ is the density of the ink, a is the characteristic length (in this case the nozzle diameter), γ refers to the surface tension, and μ is the viscosity of ink.^[10] The value of Z varied from 14.2 to 39.9 with the increasing ratio of octane. Correspondingly, TQ-inks with the above ratios leaving the nozzle are showed in Figure S2 (Supporting Information) and we can see that the stable droplet ejection occurs when Z is 16.5 (for TQ-ink of 1-cyclohexyl-ethanol: octane: *n*-butyl acetate is 6:3:1). This specific composition for the TQ-ink system is donated as TQ-ink-1. The Z value of the conventional control ink (9:1 ratio of CHB and decane) is calculated as 16.4, which is similar to that of TQ-ink-1, indicating that these two inks can eject similar sized droplets with a little adjustment of the driving voltage (Figure S3, Supporting Information).

The effect of ink on the TFB layer was examined by comparing the UV-vis absorption spectra of TFB films before and after rinsing with TQ-ink and control ink solvent (Figure 1a). Visibly, the absorption spectrum of TFB hardly changed after rinsing with TQ-ink-1, it shows barely any damage to the TFB layer. While the absorption peak vanishes after rinsing with control ink solvent, indicating serious corrosion to the TFB. Besides, the enhanced absorption spectrum from 430 to 700 nm of the layer indicates that the TFB layer becomes roughened due to control ink rinse. This rough surface of TFB leads to high scattering, and hence increased absorption of light.^[34] The unaffected and flat HTL layer after TQ-ink printing is very important for the uniform QD layer deposition on top and lays the foundation for obtaining high performance of QLED.

When it is guaranteed that a single droplet can drop homogeneously, two independent processes of the impact and the evaporation of the droplet must be considered. Because for the same ink, the driving voltage and the size of the nozzle have no change, the impact processes are similar for every droplet. Thus, the evaporation process determines the quality of the printed QD layer for the same ink. First, we characterized the contact angle of the TQ-ink on the TFB layer, showing that all compositions demonstrated small contact angles on the TFB

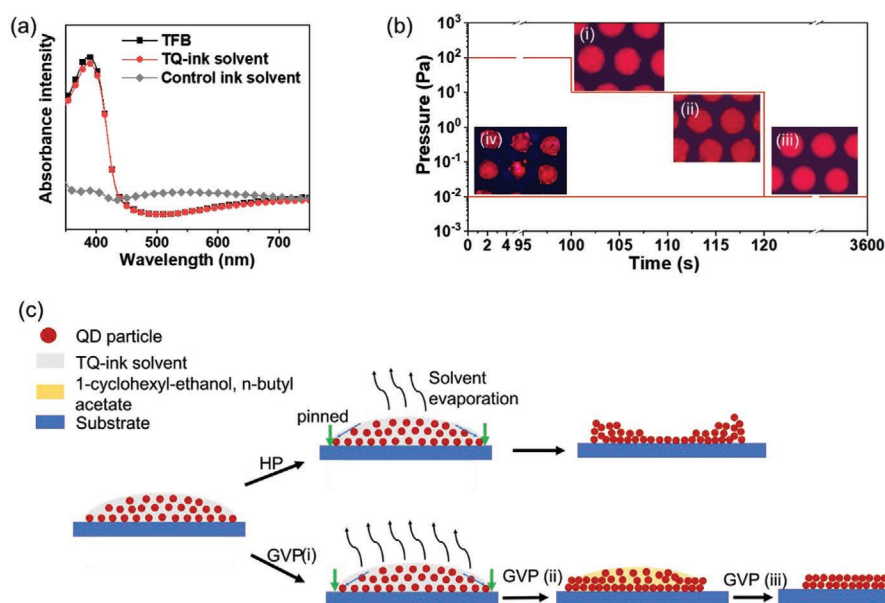


Figure 1. a) The UV-vis absorbance of the HTL layer before and after rinsing with two types of ink solvent. b) Optical microscopic images showing some typical droplet morphologies dried with different GVP stages. c) The scheme of droplet evaporation processes for HP and GVP methods.

layer varying between 3.4° and 5.3° resulting from the low surface tension of the ink (Figure S4, Supporting Information). The droplet with a small contact angle ($\approx 5.0^\circ$) is more widely spread on TFB, which can help to weaken the coffee ring effect. Because the fast evaporation rate under the large surface area suppress the particle to flow to the edge.^[28]

Besides, to cooperate with the TQ-ink system, we purpose a three-staged GVP to evaporate the solvent and form a pristine QD layer, and the conventional heat post-treatment (HP) serves as the control method. For the GVP system, a computer control program was developed to control the pumps (a molecular and a mechanical pump) connected to the vacuum chamber. Through the program, slow step-wise pressure control is adopted to evaporate three ink components successively in the chamber. As shown in Figure 1b, the GVP method contains three steps: 1) reduce the chamber pressure to 100 Pa for 100 s to pin the boundary of droplet; 2) continuously reduce the pressure to 1 Pa for 20 s for rapidly removing most of the solvent; 3) reduce the pressure to 0.01 Pa for 58 min to ensure all the solvent evaporated. As shown in Figure 1b(i),(ii), if the process ended at the first two stages, the profiles of dots present grainy, because the ultralow vapor pressure solvent (1-cyclohexyl-ethanol) are residual. When the three stages of evacuation finish, the three ink components can be removed completely, and the dots present a uniform and smooth surface (Figure 1b(iii)). However, the dots will blast and losing the original shape if the solvents only evaporated at a constant high vacuum pressure of 0.01 Pa initially (Figure 1b(iv)). This phenomenon is attributed to octane and *n*-butyl acetate with higher vapor pressures at room temperature will evaporate sharply and escapes from the ink droplet. Figure 1c illustrates the formation of the QD layer under the GVP and HP method. For the IJP technology, the rough surface of the printed layer is mainly attributed to the coffee ring effect. In short, the coffee ring is caused by the continuous flow of QD particles toward the edge of the droplet

during the evaporation of the solvent, and they are blocked and accumulated on the pinned edge to form a ring-like structure.^[24] For the HP process, the solvent evaporates slowly due to its low vapor pressure, so the QD particles have enough time to flow to the pinned edge of the droplet. Following the accumulation of QD particles, a serious coffee ring effect occurs, causing a rough QD layer (Figure S5, Supporting Information). However, for the GVP method, the higher vapor pressure solvents (octane and some of *n*-butyl acetate) are removed rapidly at stage i), which does not provide enough time for QD particle to flow, avoiding the coffee-ring to occur. Meanwhile, it can still maintain the droplet profile with the other remaining solvents. In stages ii) and iii), the remaining solvents (1-cyclohexyl-ethanol and *n*-butyl acetate) will be removed gradually. Finally, a uniform QD layer can form without pinholes and uneven surfaces on the substrates (Figure S5a, Supporting Information), guaranteeing good device performance.

To further analyze the quality of the QD layers, four QD layers were printed using two types of inks with GVP or HP method (marked as TQ-ink-1 (GVP), TQ-ink-1 (HP), control ink (GVP), and control ink (HP), respectively). Figure 2 exhibits the corresponding 3D atomic force microscopy (AFM) images. The root-mean-square (RMS) roughness values of these four types of films are 0.92, 1.72, 9.87, and 24.49 nm, respectively. The profile of TQ-ink-1 (GVP) shows the most uniform morphology due to the well-controlled evacuation process allows for the arrangement of QD within a long time, which benefits further device fabrication. The microscopic images of printed patterns on the TFB layer are shown in Figure S5 (Supporting Information). The dots of TQ-ink-1 (GVP) and TQ-ink-1 (HP) demonstrate a more uniform surface on TFB compared with those of control ink films due to their lower surface tension (23.3 N m^{-1}) and non-interfering of the underlayer. The control ink can partially redissolve the TFB layers (Figure 1a), which leads to a high RMS of the film, as shown in Figure 2c,d. Accordingly, as the dot profiles are

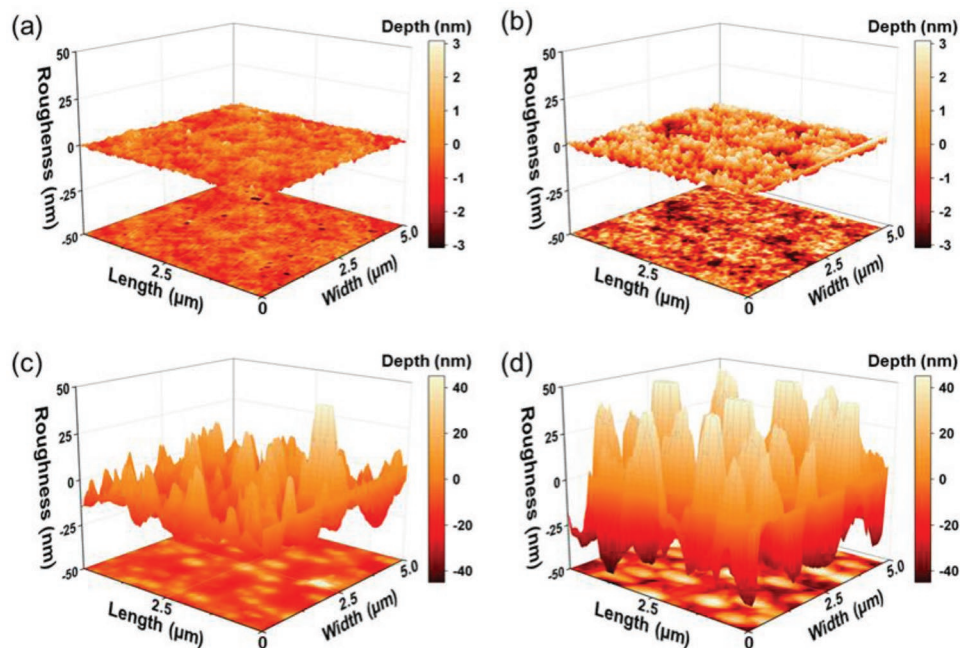


Figure 2. The AFM height 3D histograms of TQ-ink-1 dried with a) GVP and b) HP method, and of control ink dried with c) GVP and d) HP method.

shown in Figure S5a,b (Supporting Information), both patterns with two inks formed by the GVP method still demonstrate more uniform surfaces compared with those treated by HP technology. As the above discussion, because a high vacuum condition of the GVP (< 100 Pa) drives the solvent to remove more rapidly, which inhibited the QD particles to flow and accumulated at the edge.^[28] Our ink system and GVP method provide a solution to obtain the uniform and uninterference QD layer or single dot by IJP.

However, the ratio of the system needs to adjust slightly to meet different fabrication conditions. **Figure 3** shows the microscopic images of the printed dot arrays with different compositions of TQ-ink using the same GVP method. The ratio of 1-cyclohexyl-ethanol: octane: *n*-butyl acetate in the inks is 3:6:1, 6:3:1 (TQ-ink-1), 8:1:1, and pure 1-cyclohexyl-ethanol was used for comparison. It can be seen that for different ratios of the solvents, the surface profiles are different. This is because

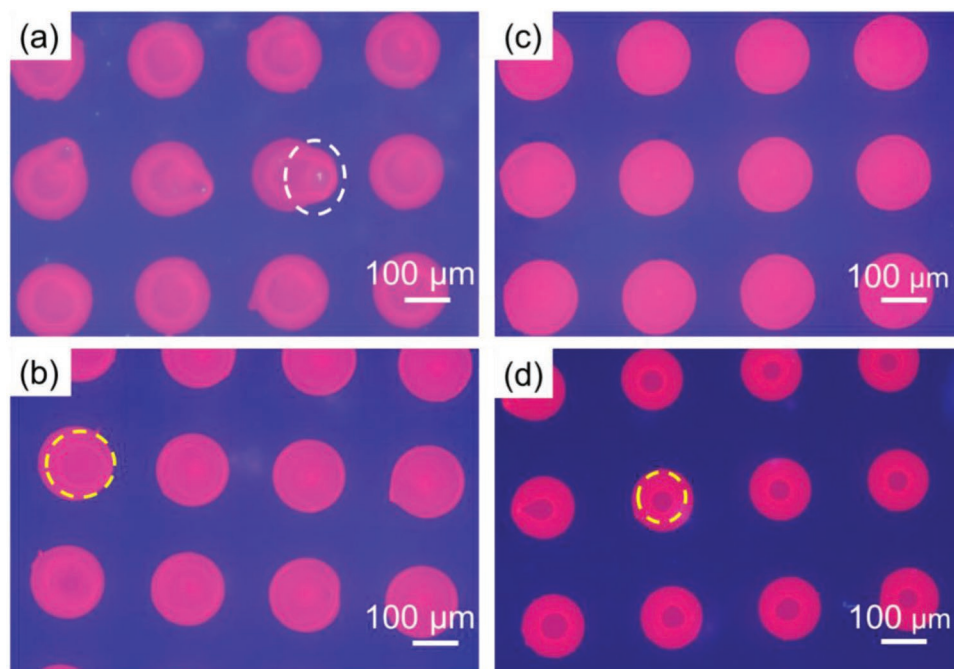


Figure 3. Optical microscopic fluorescence images of IJP dot arrays printed with different 1-cyclohexyl-ethanol: octane: *n*-butyl acetate ratios. a) 3:6:1, b) 6:3:1 (TQ-ink-1), c) 8:1:1. d) The dot array printed with pure 1-cyclohexyl-ethanol.

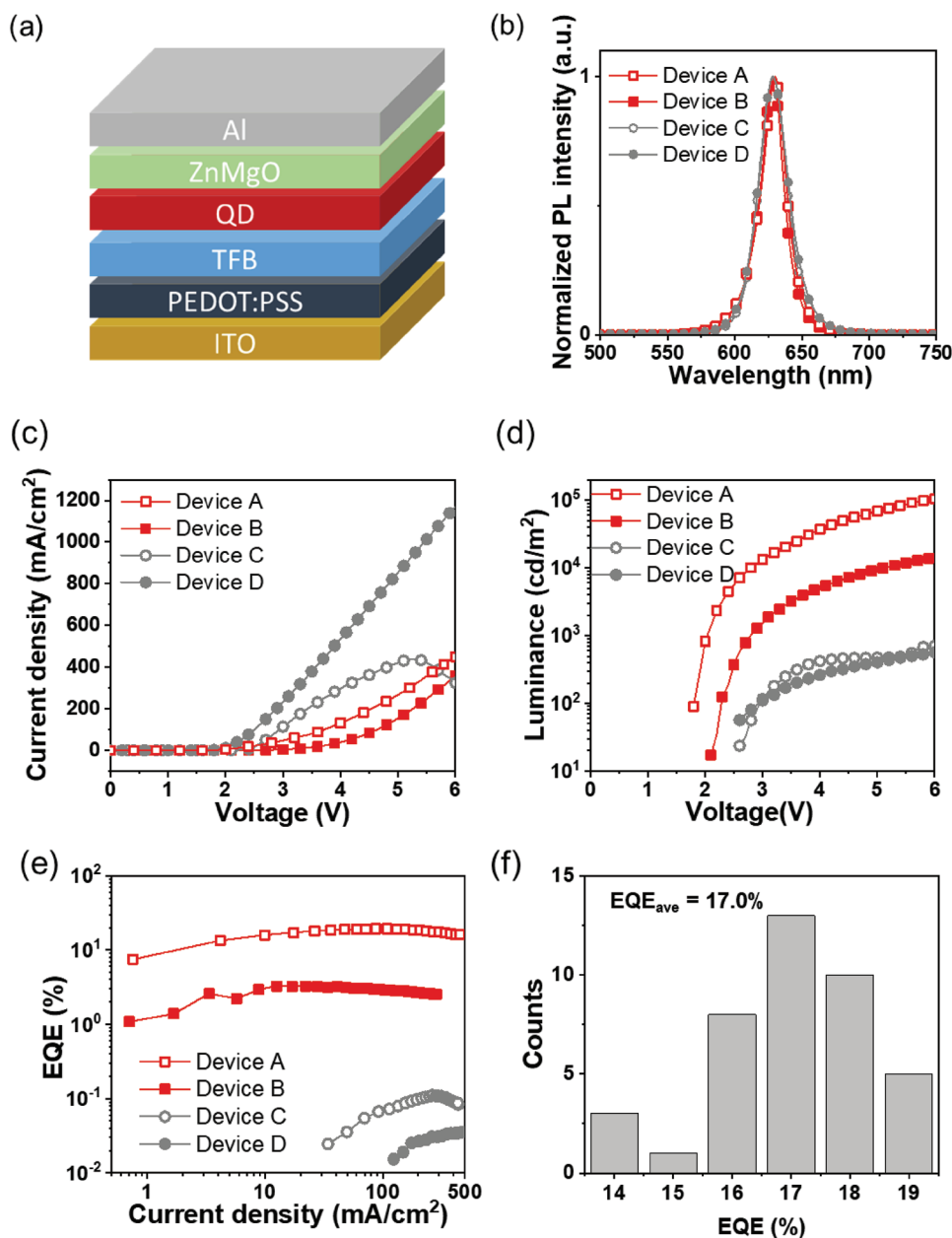


Figure 4. a) Energy band diagram of the QLED studied in this work. b) the EL spectra of TQ-ink-1 (GVP): Device A, TQ-ink-1 (HP): Device B, control ink (GVP): Device C, and control ink (HP): Device D. c) the current density–voltage curves of Device A–D. d) the luminance–voltage curves of Device A–D. e) the EQE of Device A–D. f) The statistical distribution of EQE of repeated Device A.

the evaporation speed, i.e., the vapor pressure of the mixture varies for different solvent compositions. When the ratio was 3:6:1, the printed dots show the irregular profile (white circle in Figure 3a), mainly because of the satellite droplets (Figure S2c, Supporting Information). The dots by TQ-ink-1 printed on the TFB layer (Figure 3b) maintain the even and coffee-ring free morphology. However, when the volume of 1-cyclohexyl-ethanol increased, it shows a severe coffee ring effect (yellow circles in Figure 3c,d). It can be attributed to the evaporation rate of the mixed solvent decreases, which will promote the accumulation of the QD particle at the edge of the droplet.^[17] Therefore, the composition of TQ-ink-1 shows the most suit-

able for not only the inkjet process but also for obtaining uniform QD layer.

The QLEDs were fabricated with a multilayer structure of indium tin oxide (ITO)/poly(3,4-ethylenedioxythiophene); poly(styrenesulfonate) (PEDOT:PSS)/TFB/QD/ZnMgO/Al (Figure 4a). Four types of devices include the combination of two QD Inks (TQ-ink-1 and control ink) and two post-treatment methods (GVP and HP), respectively (TQ-ink-1 (GVP): Device A, TQ-ink-1 (HP): Device B, control ink (GVP): Device C, and control ink (HP): Device D). The thickness of the IJP QD layer for device A–B are shown in Figure S6 (Supporting Information), respectively. The electroluminescence (EL) spectra show

little difference among all samples (Figure 4b), which indicates the solvents have little effect on the exciton dynamics of QDs. Figure 4c–e shows the J - V , L - V , and EQE - J characterizations, respectively. The current efficiency of the four devices and their detailed characteristics are summarized in Table S3 (Supporting Information). Device A is the champion device with the maximum luminance of 104, 826 cd m^{-2} @ 6 V and a maximum EQE of 19.3%. The luminance of Device A under the same bias is much higher than those of the other devices, especially the peak luminance of Device A is over 100 times higher than that of the control Device D. Encouragingly, the result of Device A is one of the highest EQE for IJP-QLED reported so far. As shown in Figure 4c,d, Device C and D presents very large current density, but low brightness and high turn-on voltage up to 3 V. The ideal QD layer should be uniform, compact, pinhole-free, and erosion-free to the TFB layer. Referring to our investigation, the coffee-ring effect in QD layers associated with the HP method and the underneath TFB erosion caused by the control ink leads to extensive shorting problems and large leakage currents, deteriorating the QLED device performance with very low efficiency ($EQE < 0.1\%$ of Device D). For the TQ-ink system, there is almost no interference between the QD layer and the hole transport layer (Figure 1b), and it hence guarantees a uniform distribution of current transport (no short circuit channels). Besides, the time-resolved PL (TRPL) measurement was performed to investigate the TQ- and control ink printed layers with GVP method (Figure S7 and Table S4, Supporting Information). The component (τ_1) and the average value (τ_{av}) of TQ-ink printed layer are larger than those of control ink printed one, indicating a greater number of carriers are radiatively recombined. Besides, the lifetime (τ_2) of the TQ-ink sample is larger than that of the control sample, which suggests less non-radiative recombination for the TQ-ink sample. In other words, the Auger recombination was suppressed in TQ-ink samples. Hence, both suppression of Auger recombination and leakage current led to improved efficiency of the device. For the same type of ink but adopting the HP method (Device B), the luminance @ 6V is 14,265 cd m^{-2} , and the maximum EQE is 3.26%. But they are still significantly inferior (5 times lower) compared to the GVP processed device, indicating that though the ink erosion of the TFB layer is important, the solvent drying method using GVP is crucial for the IJP-QLED device with high performance. Besides, the reproducibility of 40 IJP devices is shown in the histograms of the EQE (Figure 4e), and the average EQE is 17.0%, which suggests the fabrication of IJP QLED by using the TQ-ink system with GVP shows excellently repeatability. The results of the devices prove the importance of the uniform and no-interfere layer for the high efficiency IJP QLED, and our TQ-ink system combined with GVP is very suitable to fabricate IJP QLED.

Generally, there is always an efficiency gap between IJP and SC devices in other studies.^[9,26,30] Figure 5 compares the IJP and SC fabricated RGB QLED using the same QD materials and device structures, and the summarized data are listed in Table S5 (Supporting Information). The maximum EQE of the red, green, and blue IJP-QLEDs is 19.3%, 18.0%, and 4.4%, respectively, indicating the universality of the TQ-ink system and GVP method for all QD types. The standard red, green, and blue SC-QLEDs show the maximum EQE of 20.2%, 21.8%,

and 5.0%, respectively. The EQE of the RGB IJP-QLEDs reach 94.1%, 83.0%, and 88.0% of that of the SC-QLEDs. Our method is the first time to upgrade the performance of R/G/B IJP-QLED to the same level of SC devices by using the same QD materials and device structures, which are also the highest level among other printed QLED in the literature (Table S6, Supporting Information). That suggests that the TQ-ink system combined with GVP can reduce the loss of efficiency caused by the IJP technology, through improve the quality of layers and suppress the erosion of the HTL layer. Comparing the inkjet- and spin-coated RGB QLEDs, we can see that the blue device shows different behavior compared to the red and green ones. This is primarily due to the fact that the blue QDs are much smaller compared to the red and green QDs, thus the drying process might become different for the IJP blue QD films. However, for spin-coated QD films, the size change of the QD will not cause any difference. Besides, the large area and complicatedly patterned RGB devices fabricated by IJP are shown in the inset pictures of Figure 5, which shows the advantage of the IJP process and the feasibility of our methods.

Based on the TQ ink and GVP method, a large patterned device can be directly fabricated (Figure 6a). The enlarged microscopic image (Figure S8, Supporting Information) suggests that the as-printed QD dot pixels with a diameter of 150 μm are uniform and well-aligned. The SiO₂ pixel definition/insulation layer (200 nm thick) was fabricated on ITO,^[35] which defines the QLED pixels. The patterned RGB devices with pixel arrays are shown in Figure 6b–d, which all show homogeneous and bright luminance (@4 V). The dimension of each subpixel is 185 $\times$ 300 μm^2 , and the space between each pixel is 85 μm . Above all, the TQ-ink system and novel method extend the application of the QLED and bring the QLED tremendous potential in the future display.

In addition to the efficiency and fabrication of IJP-QLED, stability should be considered for the application of optoelectronic devices. Therefore, the operation stability of IJP-QLED was further tested under constant current stress mode. The luminance-time curves of R/G/B devices are shown in Figure 7. As shown in Figure 7, the half-life (T_{50} , the time cost while luminance degraded to 50% of the initial luminance) of the red device with the TQ-ink-1 was 312 h at an initial luminance of 11 465 cd m^{-2} , while the T_{50} of the control red device (Device C) with the same GVP method at 115 cd m^{-2} (113 mA cm^{-2}) was only 26 h (Figure S9, Supporting Information). The lifetime under different brightness was estimated by the function of $L_0^n \cdot T_{50} = C$, where C is a constant, and n is the acceleration factor. According to the previous report,^[8,14,36] $n = 1.80$ was widely adopted. Accordingly, the T_{50} of Device A is 1 588 649 h @ 100 cd m^{-2} , and 25 178 h @ 1000 cd m^{-2} , which is about five orders longer compared to Device C based on the control ink, whose lifetime is 33 h @ 1000 cd m^{-2} . Such a huge stability gap between Device A and Device C comes from the interference between QD and TFB interface. The improved morphology of the interface effectively suppressed short path and nonradiative recombination due to structural variation. On the one hand, the short path will lead to the electron and hole bypassing the QDs without exciton formation. The higher localized current density around the short path will seriously aggravate the exciton Auger process.^[21,37] Both short

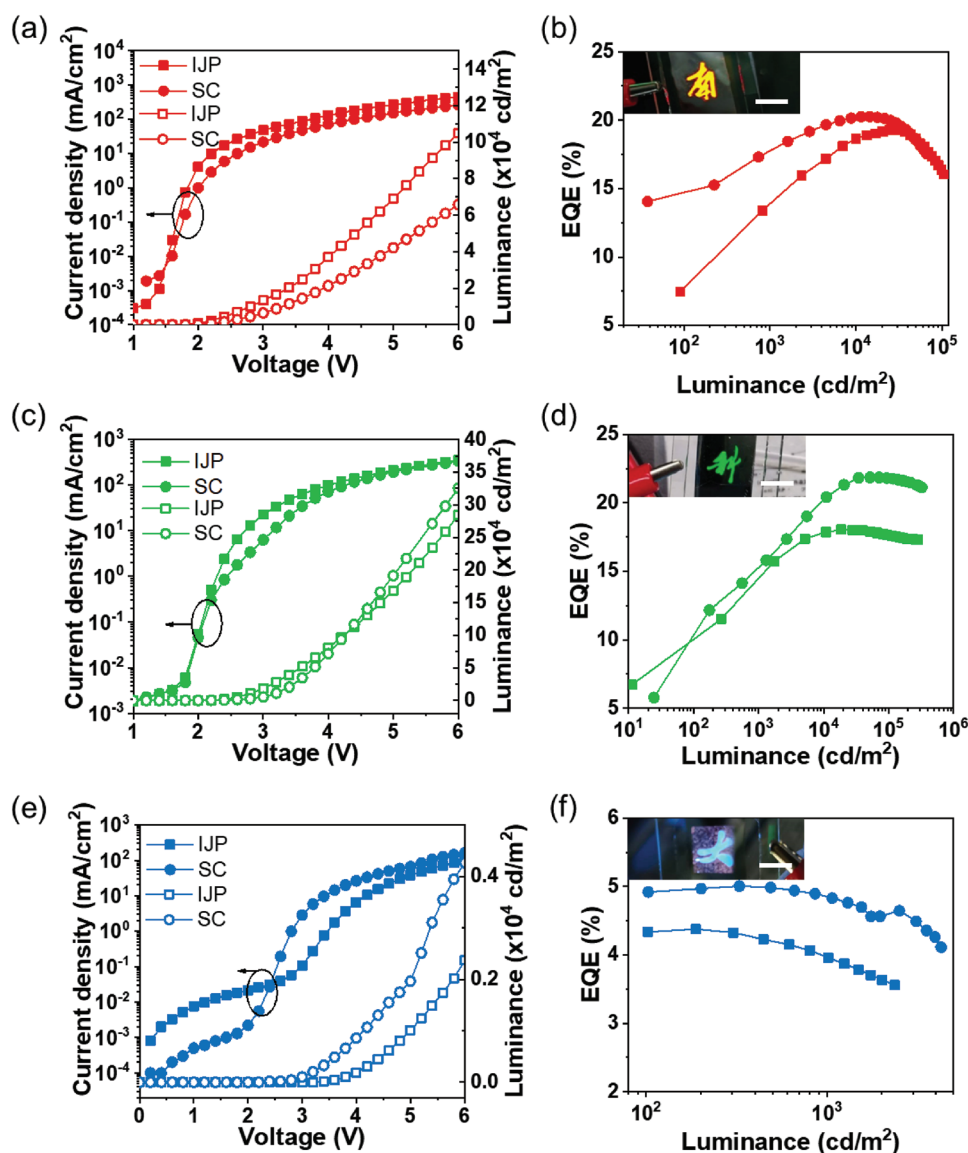


Figure 5. Current–luminance–voltage and EQE–luminance curves for IJP and SC processed red a,b), green c,d), and blue e,f) QLEDs. The RGB patterned devices by IJP are shown in the insert pictures of b,d,f), respectively (scale bar: 5 cm).

path and Auger recombination processes are catastrophic to the device stability, which will further contribute to the local heat accumulation and brush-fire.^[38] The optimized ink and GVP method prevent the architectural variation and cut off the origin of such a positive feedback degradation process, so the device stability is dramatically improved. Moreover, the green IJP QLED showed the T_{50} of 63 h @ 24 981 cd m^{-2} (Figure 7). Correspondingly, the T_{50} can be estimated as 20 655 h @ 1000 cd m^{-2} for green IJP-QLED. Besides, the T_{50} of the blue device is 46 h @ 103 cd m^{-2} (Figure 7). To the best of our knowledge, these are the best lifetime data ever reported for RGB QLED devices by IJP (Table S6, Supporting Information). Besides, the operation lifetime of the red devices by IJP and SC under constant current density were also characterized by comparing the luminance of the devices drop to 95% of their initial values (T_{95}) (Figure 7b). The red SC-QLED shows the

T_{95} of 31 h, whose initial brightness is 13 599 cd m^{-2} . And the T_{95} of IJP-QLED is over 40 h at the initial brightness of 9415 cd m^{-2} . The extrapolated T_{95} at an initial brightness of 1000 cd m^{-2} of red SC-QLED is 3401 h. The extrapolated T_{95} of IJP-QLED is 2264 h and up to 67% of SC-QLED, which is the longest T_{95} of IJP-QLED reported as far. It also results from the uniform and no-interference QD layer with the TQ-ink system and GVP method to ensure stable carrier transportation, productive exciton formation, and efficient exciton recombination. Considering the extra short lifetime with control ink, the TQ-ink system and GVP method can compensate for the loss of efficiency and lifetime caused by the IJP process. As we can see, the design of the ink system and GVP method can immensely improve the efficiency and stability of IJP-QLED by simply flat the QD layer, which can be a big step further the industry application in the future.

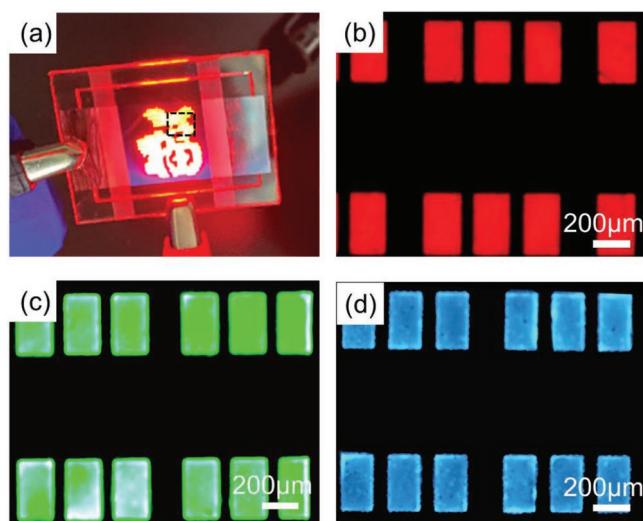


Figure 6. a) An IJP printed red QLED showing a Chinese character symboling good fortune. b–d) EL images of the IJP R, G, and B QLEDs pixel arrays under 4 V.

3. Conclusion

This work evidence that the poor performance of IJP-QLED comes from three aspects for previous research: 1) the conventional QD inks erode the HTL (TFB layer) and hence roughen the QD and HTL interface; 2) the heating post-treatment method is not suitable for the IJP process; 3) unsatisfactory surface of QD layer still occurs under one step evaporation post-treatment method. The TQ-ink system reported here can eliminate the erosion to the underlayer and the GVP technology is important in improving the surface morphology of the QD layer. The red IJP QLED exhibited an EQE of 19.3% and a long operation lifetime of 25 178 h @ 1000 cd m⁻². Moreover, the green and blue IJP-QLEDs also show outstanding EQE (18.0% and 4.4%) and operational stability (lifetime of 20 655 h @ 1000 cd m⁻² and 46 h @ 100 cd m⁻²), which are the records in literature. The methods could reduce the loss of efficiency and lifetime caused by the IJP process and be extended to other materials and devices to apply in the IJP process as well. Besides, the IJP-QLED with our method showed excellent

efficiency, stability, and high reproducibility; It is feasible to fabricate the high-resolution images and pixels easily, and it will benefit the cost-effective large-scale fabrication of high-resolution display panels.

4. Experimental Section

Material Synthesis: The red (CdZnSe/ZnS) and blue (ZnCdS/ZnS) QDs were synthesized according to ref. [39] with minor modifications. For red QD, cadmium acetate (0.16 g), zinc acetate (1.28 g), oleic acid (3 mL), and 1-octadecene (ODE) (5 mL) were mixed in a three-neck flask and evacuated under 120 °C for 1 h and filled with Ar atmosphere. An as-prepared Se precursor (2.2 M trioctylphosphine (TOP)-Se, 1.2 mL) was injected at 260 °C and reacting for 30 min. Zinc oleate and S precursor (a mixture of 2.2 M TOP-S and dodecanethiol) were injected afterward to overcoat a ZnS shell at 280 °C for 1 h. The vessel was allowed to cool down to room temperature slowly and purified with ethanol and *n*-hexane three times, and finally dispersed in mixed solvent for preparing the ink. The ink was allowed to be stabilized for 1 day before printing. For synthesizing green QD, a mixture of TOP-Se and TOP-S (molar ratio 1: 4) was injected into the Cd and Zn precursor. To obtain a blue emitting core, S precursor was altered to ODE-S (0.5 M, 2 mL) and injected to the same Cd and Zn precursor as above. The rest conditions were the same. The properties of RGB QD are demonstrated in Figure S10 (Supporting Information).

ZnMgO nanocrystals in ethanol solution were synthesized according to ref. [6] For a typical synthesis, 30 mL of a solution of zinc acetate hydrate (2.7 mmol) and magnesium acetate hydrate (0.3 mmol) in dimethyl sulfoxide and 10 mL of a solution of tetramethylammonium hydroxide (5 mmol) in ethanol were mixed and stirred for 3 h in ambient conditions, then washed twice and redispersed in absolute ethanol. The ethanol solution of ZnMgO nanocrystals (20 mg mL⁻¹) was filtered with 0.22 μm polytetrafluoroethylene (PTFE) filter before use.

Device Fabrication: QLED devices were fabricated by IJP on patterned ITO glasses (sheet resistance is about 20 Ω sq⁻¹). Before the use, all ITO glasses were cleaned in acetone, DI-water, and isopropanol for, respectively, 30 min by using an ultrasonic cleaner. Then, these were dried using nitrogen and treated in a UV-Ozone chamber for 20 min. A 30 nm of PEDOT: PSS (Lumtec.) layer was deposited and annealed at 130 °C for 20 min. Then, a 40 nm of TFB (American Dye Source, average molecular: 136 000) layer was deposited in an N₂ glove box, annealing at 100 °C for 10 min. After that, a QD layer was sequentially printed in the glovebox, and thicknesses of the QD layer was varied by changing the step length of ink drops. The post-treatment (GVP) were performed within a chamber in the same glovebox, and the chamber was vacuumed by a two-stage pumping system consisted of

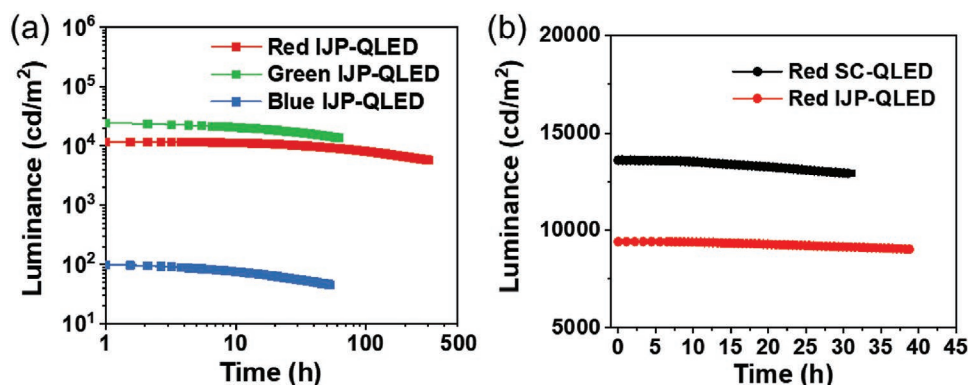


Figure 7. a) The operation lifetime of the optimized red, green, and blue IJP-QLED devices obtained using TQ-ink and GVP technologies (Red_Th @ 11 465 cd m⁻² = 312 h, Green_T₅₀ @ 24 981 cd m⁻² = 63 h and Blue_T₅₀ @ 103 cd m⁻² = 46 h). b) Stability (T₉₅) comparison of the IJP and SC fabricated red QLED devices (under a constant current stress of 45 mA cm⁻²).

a molecular and a mechanical pump. For the HP method, the devices were baking under 100 °C for 20 min. After the deposition of ZnMgO nanoparticles, the substrate was annealing at 80 °C for 20 min. The top electrodes were fabricated by heat evaporation of aluminum for 100 nm, followed by packaging using UV-glue in the glove box. The inkjet printing was performed by using a Microfab JETLAB 2 printer equipped with a piezoelectric-driven inkjet nozzle (diameter: 40 µm) in glovebox under the driving voltage (Figure S3d, Supporting Information), and the volume of a single droplet is 80 pL.

Characterizations: The surface tension and viscosity of the novel QDs inks were taken with surface tension meter (JK99B, Powereach, China) and Brookfield Rotational Viscometer (DV3T), respectively. The morphologies of the microarrays on TFB layer were characterized by using a fluorescence microscope (INSIZE ISM-D500 equipped with UV-light of 400 nm). The surface morphology of IJP QD layer was characterized by AFM (Bruker Multimode 8). The J - V - L and EQE characteristics of the devices were measured using a QLED characterization system XPQY-EQE-350-1100 (Guangzhou Xi Pu Optoelectronics Technology Co., Ltd.) combined with an integrated sphere GPS-4P-SL (Labsphere) and a photodetector S7031-1006 (Hamamatsu Photonics). The lifetime test was conducted using a lifetime test system with photodiode holders from Guangzhou Cryscos Equipment Co., Ltd. All characterization is performed under ambient conditions. SEM was performed using Zeiss Gemini SEM 300. HR-TEM was performed by using FEI Titan Themis G2 working on high-angle annular dark-field scanning TEM images. PL lifetime were recorded with by Edinburgh Instruments FS5 (excited at 450 nm and monitored at peak position) and the lifetimes were fitted by Fluoracore software.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors would like to gratefully acknowledge Y. J. Wang of Mesolight Pte Ltd. for assistance with the QD materials, Z. Chen and Z. J. Li of BOE Technology Group for assistance and support provided in carrying out the experiment as part of the joint project from the Ministry of Science and Technology of China (No. 2016YFB0401702). This work was supported by the Ministry of Science and Technology of China (Nos. 2016YFB0401702 and 2017YFE0120400), National Natural Science Foundation of China (Nos. 61674074, 61875082, 61405089, and 62005115), Key-Area Research and Development Program of Guangdong Province (Nos. 2019B010925001 and 2019B010924001), Guangdong University Key Laboratory for Advanced Quantum Dot Displays and Lighting (No. 2017KSYS007), Distinguished Young Scholar of Natural Science Foundation of Guangdong (No. 2017B030306010), Shenzhen Key Laboratory for Advanced Quantum Dot Displays and Lighting (No. ZDSYS201707281632549), Shenzhen Peacock Team Project (No. KQTD2016030111203005).

Author Contributions

S.J., H.T., and J.M. contributed equally to this work. S.Q.J. conceived the idea and conducted the ink synthesis, IJP device fabrication, characterizations, and data collection for the paper. H.D.T. conducted the devices fabrication, QLED characterizations and data collection for the paper, and J.R.M. conducted the pixelated SiO₂ and RGB SC-QLED fabrication. S.H.D. and X.W.Q. conducted the AFM characterizations. All authors discussed the design and results of the experiment. S.Q.J. wrote the paper. X.W.S., P.L., and K.W. provided suggestions and revisions. X.W.S. supervised the whole work and QLED research.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

device stability, ink system, inkjet printing, quantum dot light emitting diodes

Received: May 31, 2021

Revised: July 20, 2021

Published online: September 12, 2021

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